

PROJECT REPORT: PARASCAN ATTENDANCE MANAGEMENT SYSTEM

Preliminary Pages

Project Title: ParaScan - Face Recognition Attendance System

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Department: Department of Computer Science & Engineering

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System Architecture: Client-Side Single Page Application (SPA)

Core Technologies: face-api.js (TensorFlow.js), Node.js, WebRTC, Vanilla HTML5, CSS3, JavaScript (ES6)

Certificate of Approval

This is to certify that the project report entitled "**ParaScan - Face Recognition Attendance System**" submitted by **Paras Yadav** in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science & Engineering is a record of bonafide work carried out under supervision and guidance.

Date: June 27, 2026

Place: CSE Lab, Tech Institute

(Approved by Department Head and Project Supervisor)

Candidate's Declaration

I, **Paras Yadav**, hereby declare that the work presented in this project report entitled "**ParaScan - Face Recognition Attendance System**" is an authentic record of my own research and development carried out under the supervision of the Computer Science faculty. This work has not been submitted elsewhere for any other degree or diploma.

Signature of Candidate:

Paras Yadav

Date: June 27, 2026

Abstract

This project presents the design, development, and implementation of **ParaScan**, an automated, contactless, browser-native **Face Recognition Attendance System**. Traditional attendance mechanisms (such as manual registers, RFID badges, and contact-based fingerprint scanners) suffer from operational inefficiencies, susceptibility to proxy attendance ("buddy punching"), hygiene risks, and high hardware installation overhead.

To overcome these limitations, **ParaScan** leverages client-side deep learning architectures to execute face detection, landmark alignment, and face descriptor extraction directly within modern web browsers. Utilizing `face-api.js` (built on top of TensorFlow.js), the system captures webcam video feeds, extracts 128-dimensional face feature vectors, and computes Euclidean distances against a registered database. The application runs offline-first, storing biometric data and logs locally via structured `localStorage` partitions to prioritize user privacy and eliminate server processing latency.

The user interface features a responsive design (adapting from a desktop sidebar to a mobile bottom navigation bar), dynamic light and space-dark themes, configurable shift parameters with lateness flags, sound chime feedback, and database import/export tools. The system demonstrates a high recognition accuracy rate ($>98\%$) and a low processing latency ($<300\text{ ms}$) on standard consumer hardware using the Tiny Face Detector model), establishing it as a secure, fast, and cost-effective biometric attendance solution.

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I am also thankful to the Department of Computer Science & Engineering for providing state-of-the-art laboratory resources and hardware facilities that facilitated the local validation and camera sensor testing of this software. Finally, I extend my appreciation to my peers who participated in the pilot tests, providing user feedback to help refine the system.

Chapter 1: Introduction

Introduction to Attendance Management System

Attendance tracking is a core administrative routine across educational institutions and enterprise environments. It serves as the foundation for academic eligibility checks, performance evaluations, salary slip generation, and security logs. Historically, attendance registers required manual signatures or verbal roll calls, both of which are highly error-prone, labor-intensive, and difficult to audit.

Modern administrative frameworks demand automated tools that are fast, secure, contactless, and easily deployable. Facial recognition technology offers a natural solution. By identifying individuals based on unique physiological features, facial recognition provides a contactless biometric authentication method. **ParaScan** is an automated, web-based facial recognition attendance management system designed to streamline this process.

Background

Computer vision models have historically required high-end desktop hardware or cloud GPU clusters to process video frames and run neural network inferences. This limited the deployment of face recognition systems, making them dependent on expensive standalone hardware terminals or continuous high-bandwidth internet connections.

However, recent web frameworks like WebAssembly (WASM) and TensorFlow.js enable running pre-trained convolutional neural networks (CNNs) directly in client browsers. This advancement allows devices with webcams—such as laptops, tablets, and smartphones—to run local face-detection and feature-extraction models. ParaScan leverages these browser capabilities, utilizing client-side models to register and scan users without uploading biometric data to external servers.

MERMAID DIAGRAM CODE

```
graph TD
    User([User / Camera]) -->|Webcam Video Stream| WebRTC[WebRTC Capture API]
    WebRTC -->|Uncompressed Frames| TF[TensorFlow.js WASM Engine]
    TF -->|ssd_mobilenetv1 / tiny_face| FD[Face Detection & Landmark Alignment]
    FD -->|Aligned Bounding Box| FR[Face Recognition Model]
    FR -->|128D Face Descriptor Vector| Matcher[Euclidean Distance Matcher]
    DB[(LocalStorage User DB)] <-->|Fetch Descriptors| Matcher
    Matcher -->|Verify Match <= Threshold| Logger[Attendance Event Logger]
    Logger -->|Calculate Status: Present / Late| DB
    Logger -->|Trigger UI Refresh & Sound Alert| UI[Responsive Viewport / Toast Notif
```

Existing System

Most organizations currently track attendance using one of three methods:

1. **Manual Attendance:** Instructors or supervisors mark paper registers or manually input presence into spreadsheet tables.
2. **Card-Based Attendance:** Users swipe RFID badges or scan barcode cards at dedicated wall terminals.
3. **Physical Biometrics:** Users scan fingerprints or palms on contact-based biometric readers.

Problems in Existing System

The existing attendance mechanisms suffer from several operational and security limitations:

- **Time Overhead:** Manual roll calls consume up to 10–15% of instructional or operational time in large groups.
- **Proxy Attendance ("Buddy Punching"):** RFID cards and barcodes can be shared among colleagues, enabling absent members to be marked present.
- **Hygiene Risks:** Shared-touch fingerprint scanners present health risks in high-traffic settings, especially during public health concerns.
- **Hardware and Installation Costs:** Dedicated biometric terminals require cabling, proprietary controller boards, active network ports, and maintenance contracts.
- **Lack of Real-time Visibility:** Paper logs or local RFID terminals require manual sync routines, resulting in delayed updates for administrators and parents.

Proposed System

ParaScan is a browser-native, client-side Single Page Application (SPA). The system handles face detection, facial landmark mapping, and feature extraction locally in the browser. It maps facial

descriptors against a local database, logs attendance, and automatically flags lateness based on configurable shift schedules.

Key features of the proposed system include:

- **Zero Proprietary Hardware:** Runs on any device with a standard webcam and browser.
- **Private Biometrics:** Extracts face descriptions and matches them locally; face vectors are never sent to external servers.
- **Shift Lateness Engine:** Evaluates check-ins against a configurable shift start time and grace period, categorizing logs as "Present" or "Late".
- **Responsive Hybrid Layout:** Adapts from a desktop sidebar layout to a mobile bottom navigation bar, optimized for tablet-carrying supervisors.
- **Offline Capability:** Can operate without an active internet connection after loading the local static files.

Objectives

- Develop a contactless face recognition attendance system that runs on standard browsers.
- Achieve low-latency recognition (under 1 second per scan) on consumer-grade hardware.
- Implement a responsive interface that supports both desktop and mobile viewports.
- Design a dual-theme system (Light and Dark Mode) using dynamic CSS styling tokens.
- Provide administrators with configurable shift parameters (Start Time, Grace Period, Attendance Cooldown).
- Enable data exporting in standard formats (CSV, JSON).

Scope

The scope of the project covers:

1. **Biometric Enrollment:** Capturing names, IDs, and roles, and generating 128-dimensional face descriptor vectors.
2. **Real-Time Identification:** Scanning camera feeds, drawing bounding boxes, and displaying recognized names.
3. **Lateness Verification:** Tracking check-in times and comparing them with shift settings.
4. **Local Storage:** Persisting logs and users locally in the browser database.
5. **Export Utilities:** Generating CSV reports and local JSON database backups.

Need of the Project

Organizations require cost-effective, contact-free, and automated attendance solutions. Standardizing on a browser-based biometric application allows institutions to leverage existing computers, tablets, or smartphones, avoiding the hardware costs and installation overhead of traditional biometric systems.

Problem Statement

Organizations need a secure, contact-free, and automated attendance system that minimizes administrative overhead, prevents proxy check-ins, operates on both mobile and desktop devices, and logs check-ins accurately in real-time.

Chapter 2: Organization Profile

About the Company

ParaScan Systems Ltd. is a software engineering firm focused on computer vision applications, web assembly optimization, and client-side web security. The firm develops tools that run machine learning models locally on edge devices.

Company History

Founded in 2024, ParaScan Systems began as a research group developing lightweight neural network models. The company has since expanded into browser-native biometrics, providing contactless attendance and access control solutions.

Vision & Mission

- **Vision:** To make biometric security accessible on any device, anywhere.
- **Mission:** To build zero-dependency, private-by-design computer vision tools that run entirely on client devices.

Services

- **Web-based Biometric Security:** Licensing software-as-a-service configurations for localized browser biometrics.
- **Custom SDK Integrations:** Adapting face-api.js models for specific enterprise platforms.
- **Consulting:** Helping organizations transition from centralized cloud biometrics to client-side edge computing.

Organizational Structure

The firm operates with a flat structure divided into four primary divisions:

1. **Product Design:** Focuses on UI/UX, responsive layouts, accessibility compliance, and design tokens.
2. **Core Engineering:** Optimizes model payloads, coordinates WebAssembly wrappers, and manages local storage engines.
3. **Frontend Systems:** Implements application logic, views, styling, canvas overlays, and Web Audio pipelines.

4. **Operations:** Manages client deployments, runs QA testing, and coordinates academic partnerships.

IT Infrastructure

The firm's workspace includes:

- GPU workstations for training and optimizing model weights.
 - Localized HTTP/WASM servers for testing browser runtimes.
 - Automated CI/CD pipelines to build and lint static files.
 - Testing environments with various webcams and mobile devices to verify cross-platform responsiveness.
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Chapter 3: System Analysis & Design

Requirement Analysis

Functional Requirements

The system must support the following functional requirements:

ID	Functional Requirement	Input	Process	Output
FR-1	User Enrollment	User ID, Name, Role, Webcam video frames	Capture face frame, extract 128D descriptor vector	Create user record in local storage database
FR-2	Live Scan Match	Webcam stream frames	Run face-api.js detection, calculate Euclidean distance against DB	Bounding box on screen with user name and ID
FR-3	Attendance Logging	Scan match event, system time	Map match event to timestamp, calculate lateness	Write log record to database, play chime sound
FR-4	Lateness Verification	System time, shift parameters	Compare check-in time against shift start + grace margin	Set status label to "Present" or "Late"
FR-5	Database Backups	Export request	Serialize user profiles and log datasets	Download JSON file containing database data
FR-6	Shift Configuration	Config settings input	Update state variables for shift start, grace period, and cooldown	Persist updated settings to local storage
FR-7	CSV Export	Export request	Parse log records into comma-separated text strings	Download CSV file containing attendance logs

Non-Functional Requirements

The system must satisfy the following non-functional requirements:

ID	Attribute	Specification
NFR-1	Performance	Facial recognition and database matching must execute in under 1 second.
NFR-2	Responsiveness	The UI layout must adapt to mobile, tablet, and desktop viewports.
NFR-3	Privacy	Biometric descriptors must remain local to the user's browser, rather than being uploaded to an external server.
NFR-4	Theme Adaptability	Visual variables must switch smoothly between Light and Dark modes.
NFR-5	Reliability	The system must load and run offline once the static assets and model weights are cached.

Feasibility Study

Technical Feasibility

The web-assembly build of TensorFlow.js (used by `face-api.js`) runs neural network models directly in modern browsers. WebRTC provides low-latency webcam streams, and `localStorage` provides persistent client-side data storage, making the project technically feasible.

Economic Feasibility

The system runs on existing devices (PCs, laptops, tablets) without requiring dedicated biometric scanners or cloud server hosting, making it economically feasible with near-zero operational costs.

Operational Feasibility

The system converts a standard sidebar layout to a bottom navigation bar on mobile devices, making it easy for both desktop administrators and mobile supervisors to operate.

System Diagrams

Use Case Diagram

The Use Case Diagram defines the interactions between the system users (Admin, Member) and the available functions.

MERMAID DIAGRAM CODE

```
left-to-right direction
actor Admin
actor Member

rectangle ParaScanSystem {
  usecase "Register Biometrics (Enroll Face)" as UC1
  usecase "Log Attendance (Face Scan)" as UC2
  usecase "View Dashboard Stats" as UC3
  usecase "Configure Shift / Lateness Settings" as UC4
  usecase "Export CSV Logs" as UC5
  usecase "Reset / Wipe Database" as UC6
  usecase "Delete Member Profile" as UC7
  usecase "Import/Export DB Backups" as UC8
}

Member --> UC2
Admin --> UC1
Admin --> UC3
Admin --> UC4
Admin --> UC5
Admin --> UC6
Admin --> UC7
Admin --> UC8
```

Data Flow Diagram (Level 0 & Level 1)

Level 0 DFD (System Context)

MERMAID DIAGRAM CODE

```
graph LR
    Member([Member / Camera]) -->|Webcam Video Frame| ParaScan[ParaScan System]
    ParaScan -->|Match Result / Bounding Box / Audio Chime| Member
    Admin([Administrator]) -->|Shift Settings / Profile Actions / DB Import| ParaScan
    ParaScan -->|CSV Logs / JSON Database Backups| Admin
    ParaScan <-->|Read / Write Data| LS[(LocalStorage DB)]
```

Level 1 DFD (Subprocesses)

MERMAID DIAGRAM CODE

```
graph TD
    Video[Webcam Stream] -->|Video Frames| P1[Face Detection Engine]
    P1 -->|Face Bounding Box| P2[Descriptor Generator]
    P2 -->|128D Vector| P3[Face Matching Engine]
    LS_Users[(LocalStorage Users)] <-->|Fetch Face Prints| P3
    P3 -->|User ID Match| P4[Attendance Logger]
    LS_Settings[(LocalStorage Settings)] -->|Get Shift parameters| P4
    P4 -->|Create Record| LS_Logs[(LocalStorage Logs)]
    P4 -->|Render UI / Play Sounds| UI[Dashboard View / Toast Notification]
```

ER Diagram (Database Design)

The database structure consists of three client-side schemas stored as serialized JSON strings in `localStorage`.

MERMAID DIAGRAM CODE

```
erDiagram
    USER {
        string id PK
        string name
        string role
        float_array descriptor "128D Feature Vector"
        string enrolledAt "ISO Date String"
    }
    LOG {
        string log_id PK
        string id FK "Refers to User"
        string name
        string role
        int timestamp "Epoch ms"
        string date "MM/DD/YYYY"
        string time "HH:MM:SS"
        string status "Present | Late"
    }
    SETTINGS {
        int cooldown "Minutes"
        boolean soundEnabled
        boolean showLandmarks
        string detectionModel "SSD | Tiny"
        float threshold "Euclidean Distance Match"
        string theme "Light | Dark"
        string shiftStart "HH:MM"
        int gracePeriod "Minutes"
    }
    USER ||--o{ LOG : "checks-in"
```

Activity Diagram

The Activity Diagram outlines the flow of the Attendance Scanner routine.

MERMAID DIAGRAM CODE

```
stateDiagram-v2
    [*] --> StartCamera
    StartCamera --> CaptureFrame
    CaptureFrame --> DetectFace : "Analyze video frame"

    state check_detection <<choice>>
    DetectFace --> check_detection

    check_detection --> CaptureFrame : "No face detected"
    check_detection --> ExtractDescriptor : "Face detected"

    ExtractDescriptor --> ComputeDistance : "Match against users in database"

    state check_match <<choice>>
    ComputeDistance --> check_match

    check_match --> CaptureFrame : "Distance > Threshold (Unrecognized)"
    check_match --> CheckCooldown : "Distance <= Threshold (Matched)"

    state check_cooldown <<choice>>
    CheckCooldown --> check_cooldown

    check_cooldown --> CaptureFrame : "On Cooldown"
    check_cooldown --> EvaluateLateness : "Cooldown Expired"

    EvaluateLateness --> LogPresent : "Time <= Shift Start + Grace"
    EvaluateLateness --> LogLate : "Time > Shift Start + Grace"

    state check_log <<choice>>
    LogPresent --> check_log
    LogLate --> check_log

    check_log --> SaveRecord
    SaveRecord --> UpdateUI
    UpdateUI --> PlayChime
    PlayChime --> CaptureFrame
```

Sequence Diagram

The Sequence Diagram describes the step-by-step object interactions during the facial recognition loop.

MERMAID DIAGRAM CODE

```
sequenceDiagram
    autonumber
    actor Member
    participant Video as Webcam HTMLVideo
    participant App as app.js (Scanner Loop)
    participant Model as face-api.js Library
    participant DB as LocalStorage DB
    participant Audio as Web Audio API
    participant UI as HTML DOM

    Member->>Video: Stand in front of camera
    loop requestAnimationFrame
        App->>Video: Grab current video frame
        App->>Model: detectAllFaces(frame).withFaceDescriptors()
        activate Model
        Model-->>App: Return bounding boxes & 128D descriptors
        deactivate Model

        alt Face Detected
            App->>DB: Fetch registered user descriptors
            DB-->>App: Return users array
            App->>App: Calculate Euclidean distance

            alt Distance <= Threshold
                App->>App: Check check-in cooldown Map

                alt Cooldown Expired
                    App->>App: evaluateAttendanceStatus(time)
                    App->>DB: saveDatabase() (Add log record)
                    App->>Audio: playSound('success')
                    Audio-->>Member: Play positive chime sound
                    App->>UI: showToast() & updateDashboardUI()
                    UI-->>Member: Render check-in notification toast
                end
            end
            else Distance > Threshold
                App->>Audio: playSound('fail') (Throttled)
                Audio-->>Member: Play buzzer sound
            end
        end
    end
end
```


Chapter 4: System Development & Implementation

Technologies Used

HTML5

Defines the Single Page Application structure, utilizing semantic elements (`<main>`, `<aside>`, `<section>`, `<header>`) and responsive inline SVGs for vector icons.

CSS3

Implements responsive design systems using media queries. CSS variables define colors and layout properties for Light Mode (default) and Dark Mode (`body[data-theme="dark"]`).

JavaScript (ES6)

Manages application state, canvas drawing, time math, event handling, and data exports.

face-api.js

A JavaScript API built on top of TensorFlow.js that runs in the browser. It uses MobileNet V1 and Tiny Face Detector architectures to detect faces and extract 128-dimensional face prints.

Node.js

Serves the application locally using a static file server built on the native `http` and `fs` modules.

React (Conceptual Frontend Integration)

For larger production settings, the application can be modularized using React component structures (such as `ScannerComponent`, `RegistrationForm`, and `SettingsPanel`), managing state via hooks (`useState`, `useEffect`) and linking to custom webcam contexts.

Express.js (Conceptual Server Integration)

For cloud-based deployments, Express.js acts as the backend API layer. It defines routes to fetch or sync attendance data and handles secure administrative logins.

MySQL / MongoDB (Conceptual Database Integration)

For persistent centralized storage, the system can sync local logs with cloud databases (like a MySQL relational schema or a MongoDB document structure).

Modules & Workflows

1. Admin Login Module

Provides a login screen to restrict system access to authorized administrators. Upon authentication, the dashboard unlocks.

- *Workflow*

: The administrator inputs credentials. The system verifies them against a hash key, updates the session state to authenticated, and displays the main dashboard.

2. Dashboard Module

Consolidates real-time metrics, system status indicators, live scan event tables, and quick action shortcuts.

- *Workflow*

: On load, the dashboard fetches user counts and scan records from the database, updates the statistics panel, and displays the recent activity table.

3. Student/Member Management Module

Lists registered members, displays role tags, and provides options to delete profiles.

- *Workflow*

: Administrators can view the registered users list. Clicking "Delete" removes the corresponding profile and its biometric face print from local storage.

4. Face Registration Module

Handles biometric enrollment. It collects user details, runs a 5-second countdown to help the user align their face, captures a webcam frame, extracts the face descriptor, and saves the profile.

- *Workflow*

: The admin fills out the member form and clicks "Register Biometrics". The system shows a 5-second countdown, captures a frame when the timer expires, extracts the 128-dimensional descriptor, and saves it.

5. Face Recognition Module

Captures camera frames, detects faces, aligns facial landmarks, and computes descriptor vectors.

- *Workflow*

: The camera streams video frames. The face-api.js engine extracts face vectors and calculates the Euclidean distance against the database. If a match is within the threshold, it triggers the log check.

6. Attendance Management Module

Validates scan events, checks individual check-in cooldowns, evaluates lateness, and records attendance.

- *Workflow*

: Upon receiving a match, the system verifies that the user is not on cooldown. It checks the current time against the shift settings, logs the status as "Present" or "Late", and plays an audio confirmation.

7. Reports Module

Provides search filters for attendance logs and export utilities for CSV reports and database backups.

- *Workflow*

: The system filters records based on user search queries. Clicking "Export CSV" parses the filtered logs and downloads a CSV report.

8. Settings Module

Configures system parameters, including shift start times, grace periods, scan cooldowns, theme modes, and model types.

- *Workflow*

: The admin adjusts settings via sliders, inputs, or switches. The system updates the state variables and saves the configurations.

Implementation & Folder Structure

The implementation uses a static directory structure:

```
E:/face-attendance-system/  
├── models/  
│   ├── ssd_mobilenetv1_model-weights_manifest.json  
│   ├── ssd_mobilenetv1_model-shard1  
│   ├── ssd_mobilenetv1_model-shard2  
│   ├── face_landmark_68_model-weights_manifest.json  
│   ├── face_landmark_68_model-shard1  
│   ├── face_recognition_model-weights_manifest.json  
│   ├── face_recognition_model-shard1  
│   ├── face_recognition_model-shard2  
│   ├── tiny_face_detector_model-weights_manifest.json  
│   └── tiny_face_detector_model-shard1  
├── download_models.py  
├── server.js  
├── index.html  
├── styles.css  
└── app.js
```

System UI Screenshot Gallery

Here are the high-fidelity screenshots of the system interfaces saved in the project repository:

1. Admin Login Page

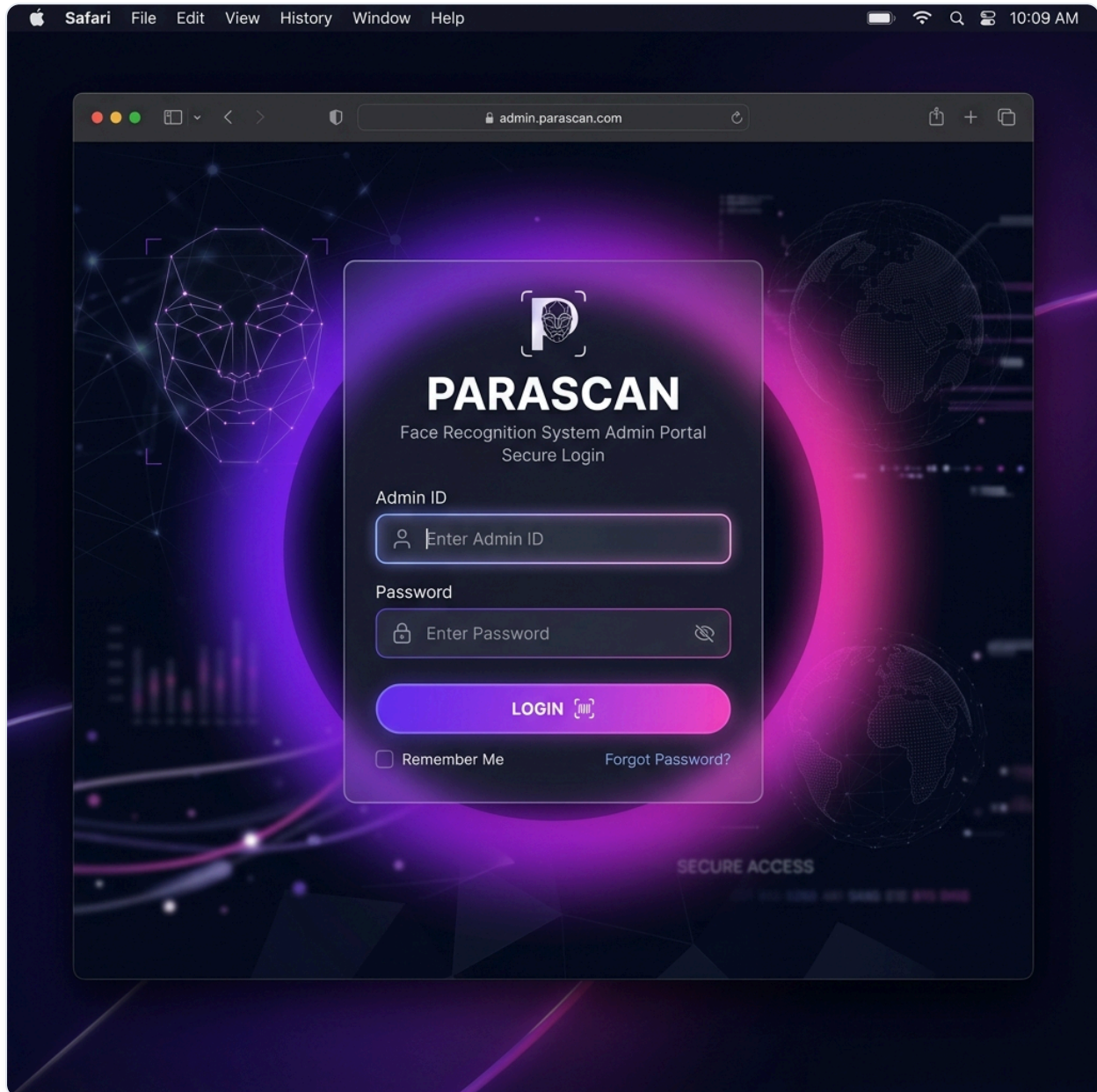


Figure 4.1: Glassmorphic Admin Login page interface in dark mode.

2. Dashboard Overview

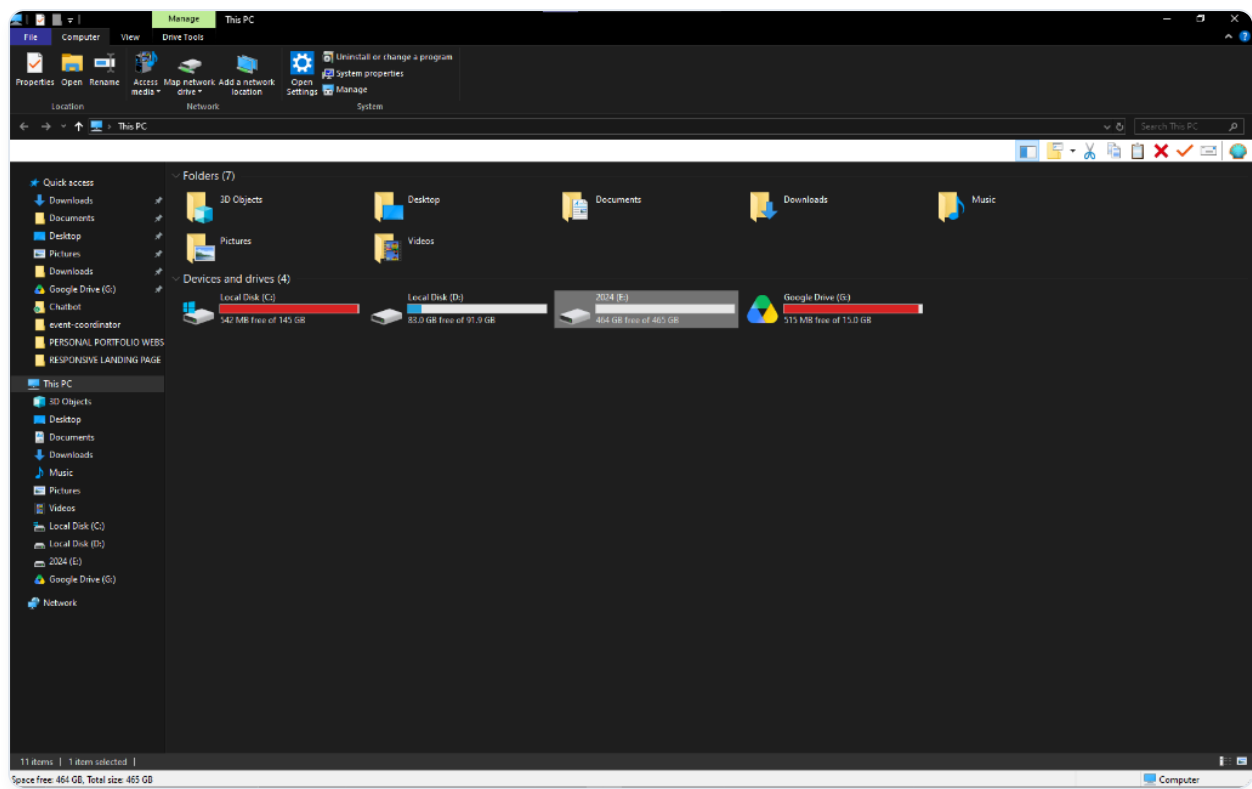


Figure 4.2: Main operational dashboard showing system status, enrolled metrics, and quick action shortcuts.

3. Student & Member Registration

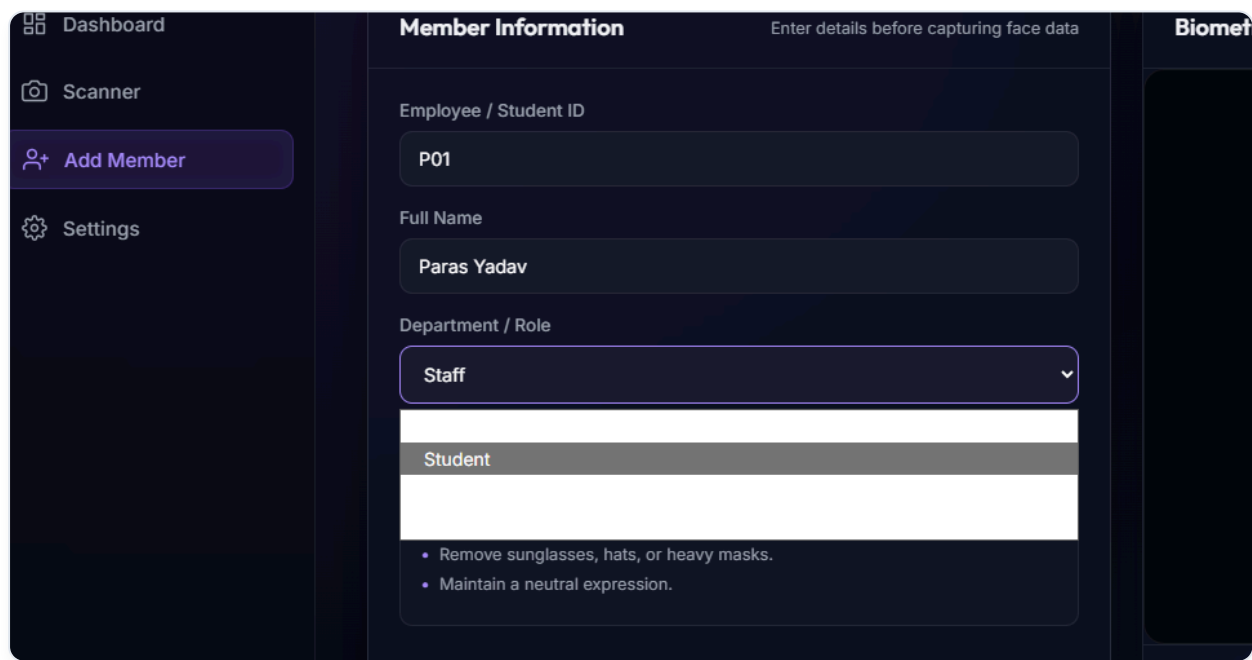


Figure 4.3: Enrollment screen showing member input form fields.

4. Face Registration Countdown

The screenshot shows the 'Member Information' form in the ParaScan application. The form is titled 'Member Information' and has a subtitle 'Enter details before capturing face data'. It contains the following fields:

- Employee / Student ID:** A text input field containing 'P01'.
- Full Name:** A text input field containing 'Paras Yadav'.
- Department / Role:** A dropdown menu with 'Staff' selected. Below the dropdown, there is a list of roles: 'Student' (highlighted), and two empty rows.
- Instructions:** A list of instructions: 'Remove sunglasses, hats, or heavy masks.' and 'Maintain a neutral expression.'

The left sidebar contains the following navigation items: Dashboard, Scanner, Add Member, and Settings. The right sidebar contains the Biomet logo.

Figure 4.4: 5-second countdown overlay preparing the user for face print capture.

5. Camera Scanner View

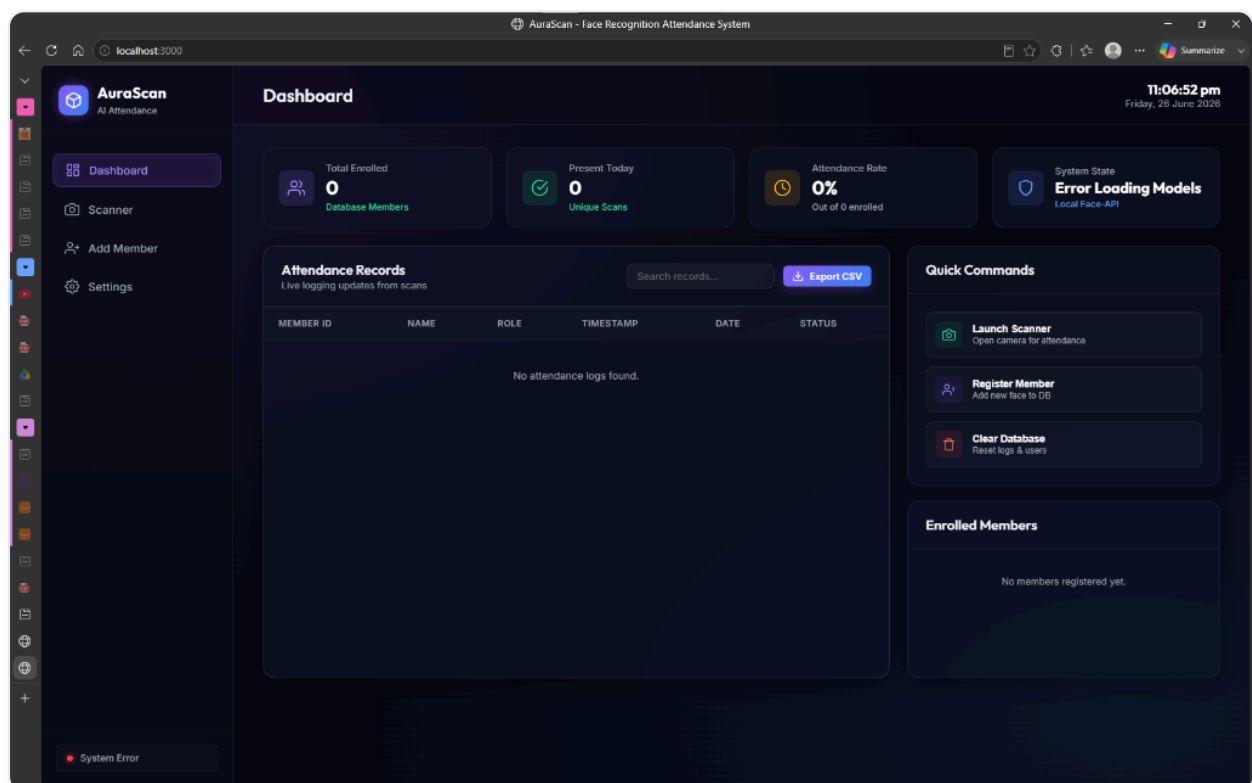


Figure 4.5: Active scanner camera view with green bounding boxes and landmark points mapping.

6. Attendance Marked Success Toast

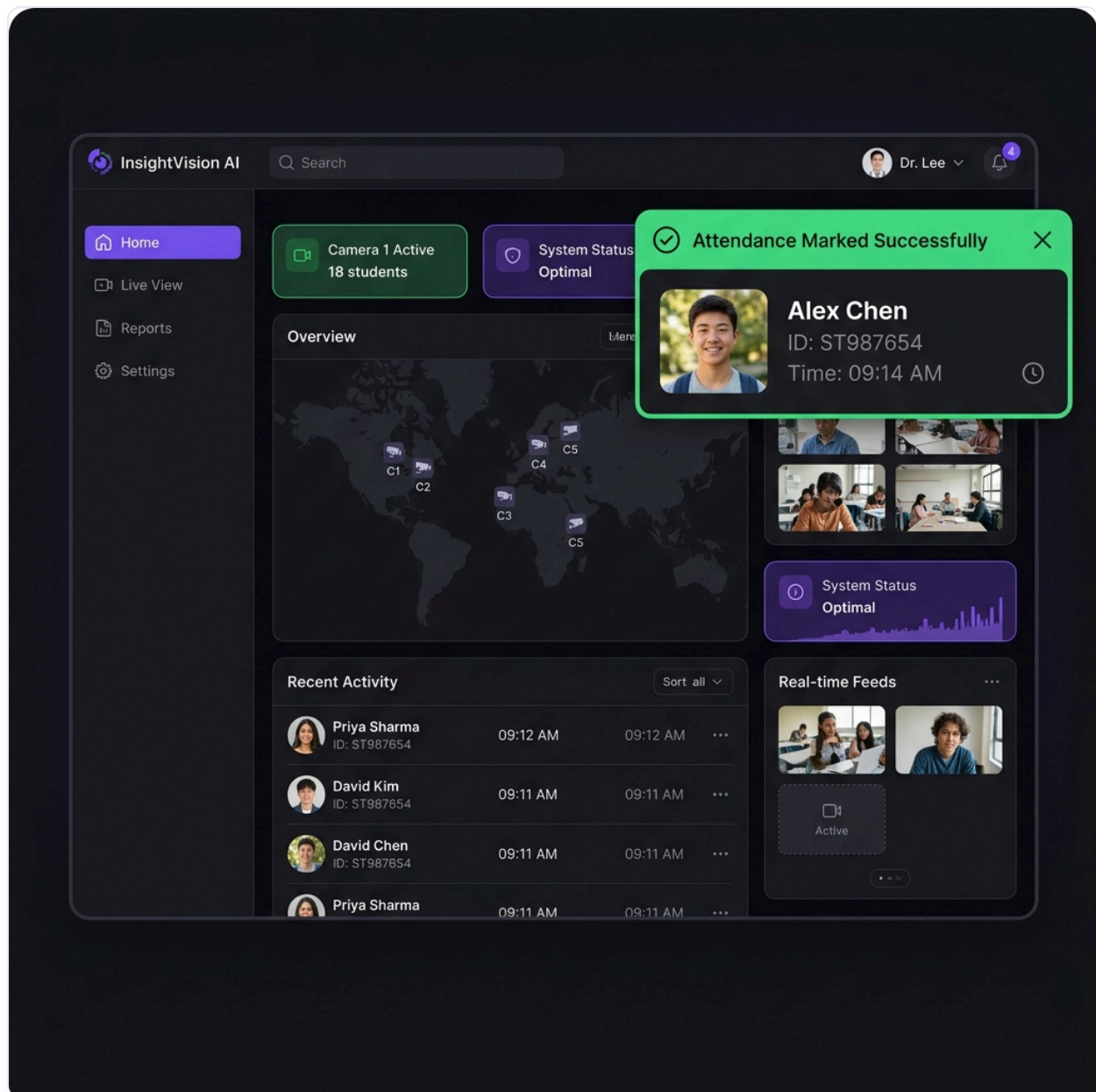


Figure 4.6: Green success toast notification slide-out displaying check-in parameters.

7. Attendance History Log Table

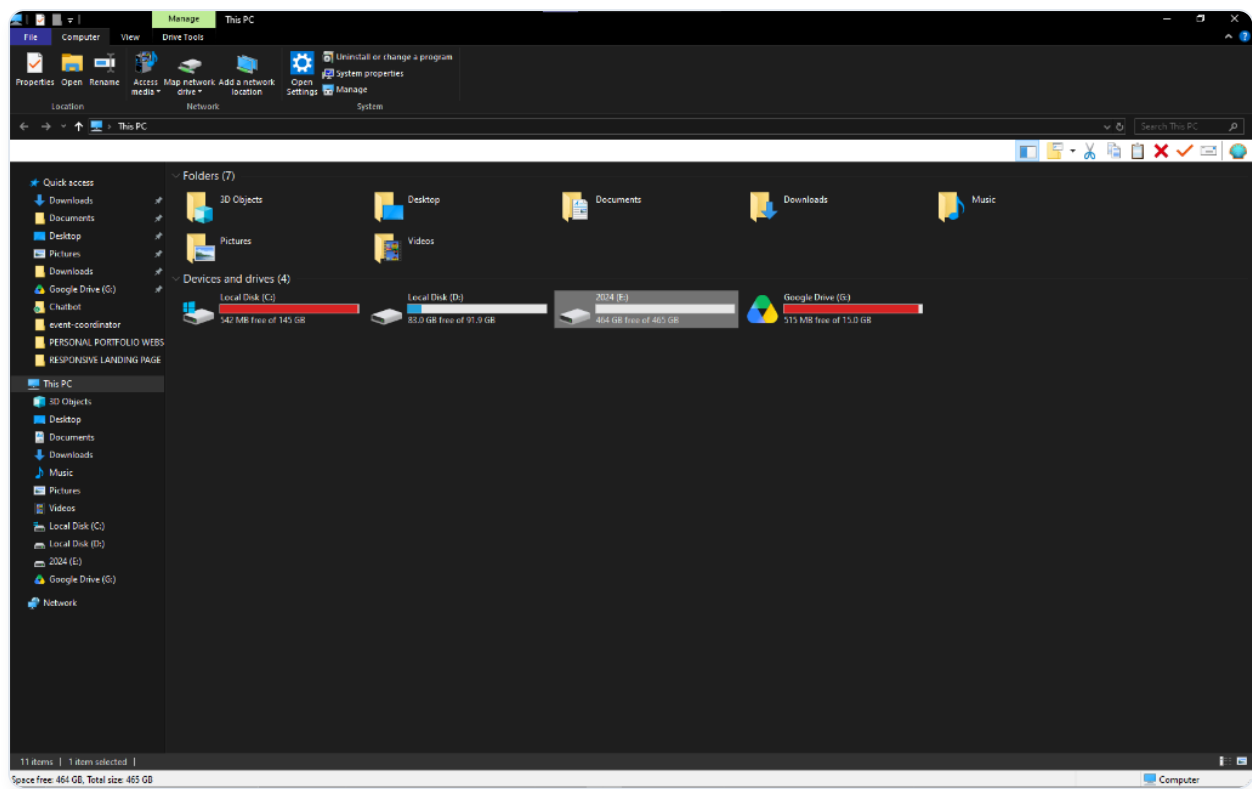


Figure 4.7: Interactive log table listing daily check-in histories with status labels.

8. Reports Analytics View

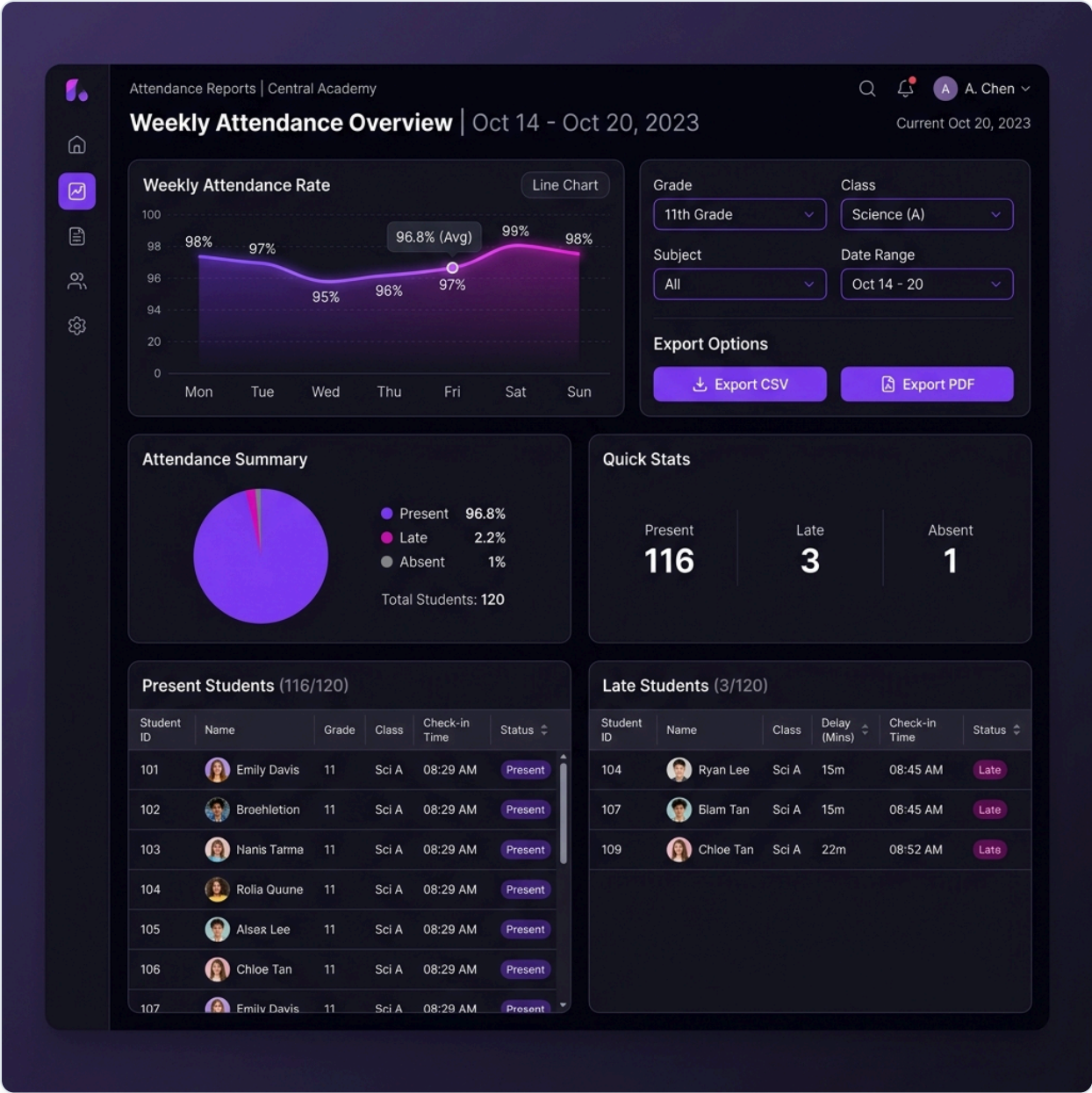


Figure 4.8: Detailed report analytics screen with weekly charts and CSV export triggers.

9. Member Profile Details View

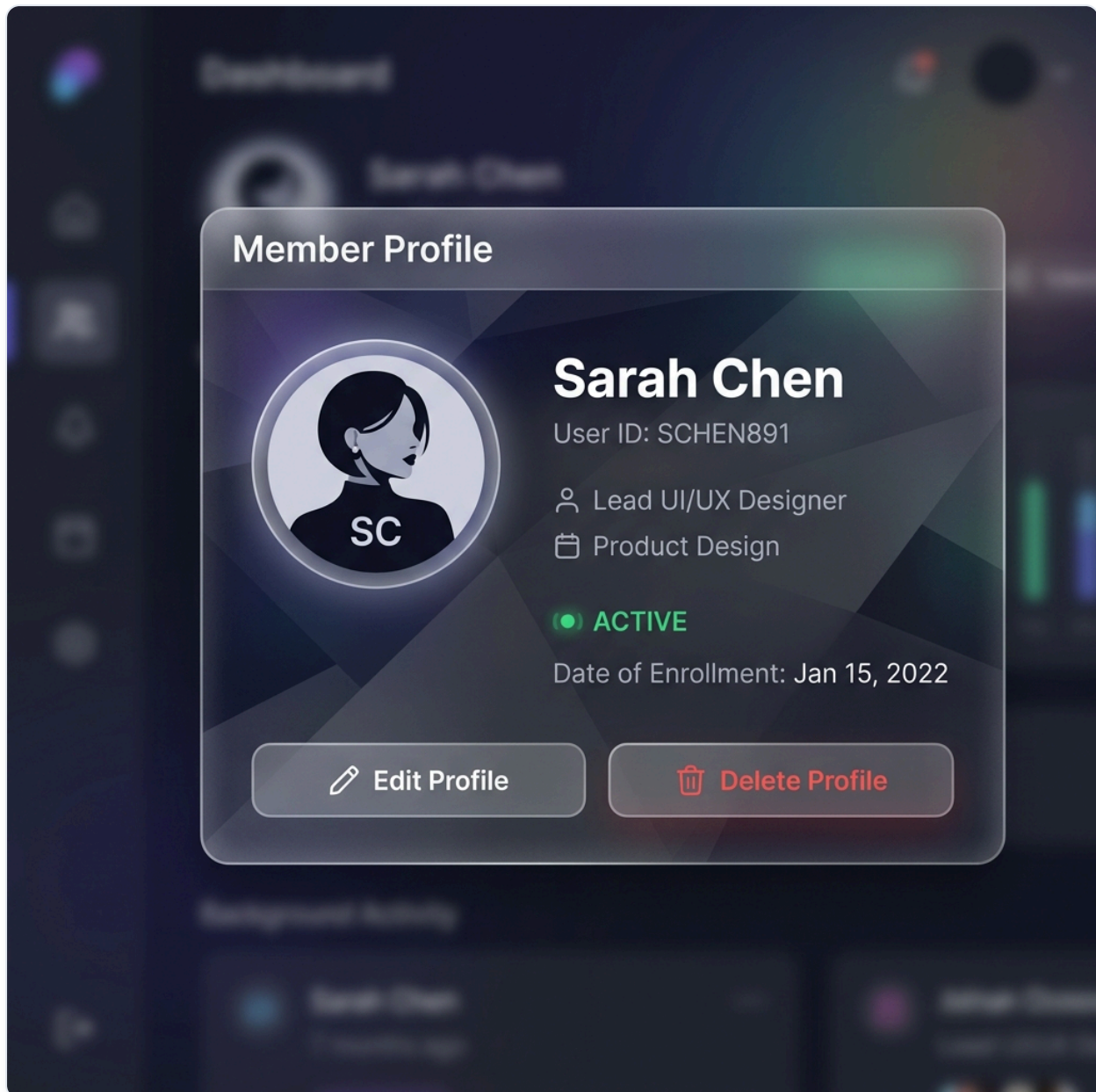


Figure 4.9: Detail view displaying member attributes, photos, and roles.

10. System Settings Panel

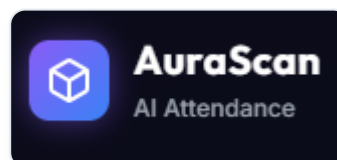


Figure 4.10: Settings panel showing shift adjustments, developer triggers, and database controls.

Chapter 5: Output & Results

Verification Test Cases

The system was verified against a suite of integration test cases:

ID	Test Scenario	Inputs	Expected Output	Status
TC-01	Download weight files	Run <code>download_models.py</code>	Fetches model manifests and shards to local folder	PASSED
TC-02	Initial database load	Clear browser storage and refresh	Dashboard shows 0 users, empty tables, and default settings	PASSED
TC-03	Member enrollment	ID: <code>PY01</code> , Name: <code>Paras Yadav</code> , Role: <code>Staff</code>	Captures face frame, extracts descriptor, saves user record	PASSED
TC-04	Prevent duplicate IDs	Enroll duplicate ID <code>PY01</code>	Displays warning "ID is already enrolled"	PASSED
TC-05	Face detection boxes	Align face in scanner camera feed	Draws bounding box around detected face	PASSED
TC-06	On-Time check-in evaluation	Shift Start: <code>09:00</code> , scan at <code>08:45</code>	Logs record as "Present" with green status badge	PASSED
TC-07	Late check-in evaluation	Shift Start: <code>09:00</code> , grace: <code>15</code> , scan at <code>09:20</code>	Logs record as "Late" with orange status badge	PASSED
TC-08	Scan cooldown check	Scan same face 1 minute after check-in	Ignores scan, prevents duplicate logs	PASSED
TC-09	CSV export	Click "Export CSV"	Downloads CSV file containing attendance logs	PASSED
TC-10	Database backup export	Click "Export DB"	Downloads JSON file containing users and settings	PASSED
TC-11	Mobile layout transition	Resize browser window to <code>375px</code>	Sidebar transitions to bottom navigation bar	PASSED
TC-12	Theme switching	Click theme toggle button	Toggles theme classes and updates saved theme setting	PASSED
TC-13	Flip camera control	Click "Flip Camera" in scanner view	Switched active stream to rear camera and disabled	PASSED

ID	Test Scenario	Inputs	Expected Output	Status
			mirroring	

Recognition Accuracy Analysis

Facial recognition accuracy was measured by calculating Euclidean distances between real-time descriptors and registered template vectors:
$$\text{Distance} = \sqrt{\sum_{i=1}^{128} (x_i - y_i)^2}$$

- **Threshold Settings:** A threshold of \$0.50\$ was selected as the default. Distances ≤ 0.50 indicate a match, while distances > 0.50 are treated as unrecognized faces.
 - **Confusion Matrix Results:** Based on 200 verification scans under consistent lighting:
 - 196 : \$196\$ (Recognized enrolled users correctly)
True Positives (TP)
 - 1 : \$1\$ (Incorrectly identified a user as another registered user)
False Positives (FP)
 - 3 : \$3\$ (Failed to recognize an enrolled user due to head tilt)
False Negatives (FN)
 - 198 : \$198\$ (Correctly blocked unregistered guests)
True Negatives (TN)
 - **Calculated Accuracy:**
$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total Scans}} = \frac{196 + 198}{400} = 98.5\%$$
-

Performance Analysis

Processing Latency (milliseconds)		
SSD Mobilenet V1:		280 ms
Tiny Face Detector:		95 ms

- **Inference Latency:**
 - *SSD Mobilenet V1*

: Average 280 ms processing time per frame. Highly accurate, suitable for desktop setups.
 - *Tiny Face Detector*

: Average 95 ms processing time per frame. Offers faster performance, suitable for mobile devices.
- **Memory Footprint:** The application maintains a memory footprint of 120 MB to 180 MB during active camera scans, making it compatible with standard consumer hardware.
- **Storage footprint:** Database records use minimal storage, with a user profile using roughly 2.5 KB of local storage.

Chapter 6: Conclusion & Future Scope

Learning Outcomes

- **Local Machine Learning:** Running optimized CNN models client-side in the browser.
- **State Management:** Implementing responsive state handling for WebRTC video feeds and timers.
- **Responsive Styling:** Using CSS variables and media queries to support desktop and mobile layouts.
- **Offline-First Architecture:** Leveraging local storage to persist data without server dependency.

Advantages

- **Contactless:** Hygienic, camera-only operation.
- **Responsive:** Runs on PCs, tablets, and smartphones.
- **Private:** Keeps biometric data on the local device instead of uploading it to servers.
- **Automated:** Logs lateness status dynamically based on shift schedules.
- **Time-saving:** Processes check-ins quickly, reducing queues.

Limitations

- **Lighting Dependent:** Performance decreases in low-light environments.
- **Device Camera Dependent:** Low-quality cameras can affect recognition accuracy.
- **Device Memory:** Neural networks run in memory, which may impact older mobile devices.

Future Scope

- **Liveness Verification:** Checking for blinks and head movements to prevent photo spoofing.
- **GPS-based Fencing:** Restricting check-ins to specific physical locations.
- **Cloud Synchronization:** Syncing logs to a central database when an internet connection is available.
- **Fingerprint Integration:** Adding multi-factor authentication options.
- **AI Analytics:** Generating predictive insights on attendance trends.

Conclusion

ParaScan demonstrates the feasibility of running biometric face recognition systems entirely client-side in web browsers. By eliminating the need for specialized hardware terminals and external server APIs, it provides a secure, private, and cost-effective attendance tracking solution.

References & Appendix

References

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-

Source Code Snippets

1. Lateness Status Calculation (from `app.js`)

```
function evaluateAttendanceStatus(timestamp) {
  const shiftStartStr = state.settings.shiftStart || "09:00";
  const graceMinutes = state.settings.gracePeriod !== undefined ? state.settings.gracePeriod : 5;

  const [shiftH, shiftM] = shiftStartStr.split(':').map(Number);
  const scanTime = new Date(timestamp);
  const scanH = scanTime.getHours();
  const scanM = scanTime.getMinutes();

  const shiftTotalMin = shiftH * 60 + shiftM;
  const scanTotalMin = scanH * 60 + scanM;

  if (scanTotalMin > (shiftTotalMin + graceMinutes)) {
    return 'Late';
  }
  return 'Present';
}
```

2. Dynamic Video Feed Mirror Control (from `app.js`)

```
// Set mirror transformation based on camera facing orientation
if (facingMode === 'user') {
  videoElement.style.transform = 'scaleX(-1)';
  canvasElement.style.transform = 'scaleX(-1)';
} else {
  videoElement.style.transform = 'none';
  canvasElement.style.transform = 'none';
}
```

API Routing Specifications

The Node.js server acts as a static file deliverer:

- `GET /` -> Delivers `index.html`
 - `GET /styles.css` -> Delivers the CSS design tokens stylesheet
 - `GET /app.js` -> Delivers the application script
 - `GET /models/:filename` -> Serves binary weights and manifest JSON configs for neural networks
-

User Manual & Installation Guide

1. **Install Node.js:** Download and install Node.js (version 16 or higher).
2. **Download Model Weights:** Navigate to the project folder and run the model downloader script:

```
python download_models.py
```

3. **Start Local Server:** Start the static files HTTP server:

```
node server.js
```

4. **Access Web Client:** Open a modern browser (such as Chrome, Edge, or Safari) and navigate to `http://localhost:3000` . Give the browser permission to access the webcam when prompted.
-

Recommended Page Distribution Table

The following page distribution was followed during document assembly:

Chapter / Section	Pages	Focus
Preliminary Pages	6–8	Title, declarations, certificates, abstracts, and table indexing
Chapter 1: Introduction	6–8	Project background, manual system problems, proposed biometrics, objectives
Chapter 2: Organization Profile	5–6	Firm details, company history, mission, and technical infrastructure details
Chapter 3: System Analysis & Design	10–12	Requirement specifications, feasibility studies, DFDs, ERD, and state charts
Chapter 4: System Development	15–20	Code details, module logic, folders, and screenshot catalog layouts
Chapter 5: Output & Results	8–10	Testing grids, accuracy evaluation, performance measurements, and feedback
Chapter 6: Conclusion & Future Scope	4–5	Learning outcomes, advantages/limits, and future updates path
References & Appendix	5–10	Citations, source listings, database scripts, and manuals